

DEEKSHA REGUPATHI

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Portfolio : <https://deeksha-regupathi.github.io/portfolio>

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CAREER OBJECTIVE

Electronics and Communication student with a keen interest in technology, problem-solving, and learning. Have a good foundation in electronics, programming, and web development, as well as practical experience in small-scale academic projects. Looking for internships to gain experience in the real world, improve technical skills, and perform well in a global working environment.

EDUCATION

Adharsh Vidyalaya Higher Secondary School HSC (12th): 80% (2022-2023)

Park Matriculation Higher Secondary School SSLC (10th): 100% (All Pass) (2022-2021)

B.E. Electronics and Communication Engineering (2023 – Present)

- **Current CGPA: 8.45**
- **Semester Exchange:** INTI International University, Malaysia .

TECHNICAL SKILLS

- **AI / Computer Vision:** YOLO (v8/v10), OpenCV, TensorFlow, PyTorch, PySpark, Spark MLlib.
- **IoT & Hardware:** Arduino, ESP32, Sensor Fusion, Blynk, Cisco Packet Tracer, ROS.
- **Programming:** Python (Advanced), C++, Linux CLI, Git.
- **Web / UI for IoT:** HTML5, CSS3, JavaScript, Bootstrap (Dashboard Design).

SOFT SKILLS

- Project Communication
- Team Work
- Public Relations

TECHNICAL PROJECTS

Edge AI: YOLO-Based Roadside Object Detection

Developed a real-time object detection system using YOLO, covering the full pipeline from manual data collection to Roboflow-based annotation and model training. Optimized local deployment environments to reduce inference latency and integrated Python-based alert mechanisms for roadside monitoring applications.

Big Data: Spark-Powered Fraud Detection Pipeline

Engineered a scalable fraud detection pipeline using PySpark to process 284K+ imbalanced records. Implemented Random Forest and Logistic Regression with Spark MLlib, achieving 0.99 ROC-AUC and reducing false negatives by 94% through effective class imbalance handling.

Full-Stack IoT: Glacial Flood Alert System

Designed a mission-critical IoT platform for real-time environmental monitoring in glacial regions. Developed a web-based dashboard using JavaScript and Bootstrap to visualize sensor data via HTTP/MQTT and implemented threshold-based logic for immediate emergency alerts.

Smart Library Entry System

Developed using HTML, CSS, JavaScript
Role-based login for students, faculty, and staff
Helps manage and monitor library entries

IoT-Based Zone Vehicle Speed Control System

Built using ESP32 microcontroller and RFID authentication
Controls motor through motor driver after valid RFID detection
Connected to ThingSpeak cloud for real-time monitoring

IoT: Emergency Vehicle Priority & Smart Home Systems

- **Brilliant Bharat Hackathon 2025:** 1st Round Qualifier for a CV-based traffic signal automation system.
- **AtomQuest 2024:** Selected in Top Tier for an Arduino-based automation system; achieved **70% reduction** in manual operation latency for 5+ devices.

Certifications & Technical Activities

1. Course Certifications

Diploma in Computer Applications – Angel Computer Centre
Front End Development (HTML) – Great Learning Academy

2. Workshop / Training Certifications

IEEE SPS Season School –Vel tech University
IoT Prototyping with Arduino Workshop – Shaastra, IIT Madras
ROS Uncovered Robotics Workshop – PSG College of Technology

3. Hackathon / Competition Certifications

VISA 24 Hrs AI Hackathon – IIT Madras
Industrial AI Hackathon – IIT Madras
Tech Trek Hackathon Season 1 – PPG College
Brilliant Bharat Hackathon 2025 – IGCAR Kalpakkam

4. Event / Participation Certifications

Demo Day – Ethics of AI & Data Privacy – Karpagam Academy of Higher Education
AtomQuest 2024 – Atomberg Technologies
Pixel Wizard Web Development Event – Sant Longowal Institute of Engineering and Technology

PROFESSIONAL EXPERIENCE

AI Intern | NoviTech | *April 2025 – May 2025*

- Contributed to ML model evaluation and feature engineering for industrial analytics.
- Optimized data preprocessing scripts, reducing model training preparation time.

Industrial Intern | BSNL | *June 2024*

- Analyzed digital switching systems and telecom infrastructure at a regional scale.

LANGUAGES:

English (Fluent),

Mandarin (Technical Basic/Stage 1),

French (Basic)

Tamil & Telugu (Native).